



2023

COMPUTER SCIENCE — GENERAL

Paper : DSE-A-1, DSE-A-2 and DSE-A-3

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Paper : DSE-A-1

(Database Management System)

Full Marks : 50

Answer **question no. 1** and **any four** questions from the rest.

1. Answer *any five* questions :

2×5

- (a) Define Database Schema.
- (b) Name two DML commands in SQL.
- (c) What do you understand by composite attributes?
- (d) What is weak entity?
- (e) What do you understand by entity integrity?
- (f) What do you understand by domain constraints?
- (g) Define transitive dependency.
- (h) Write the problems of un-normalized database.

2. (a) State the main features of RDBMS.

(b) What is Data Definition Language?

(c) Elaborate the main roles of a database administrator.

4+2+4

3. (a) What are the different types of mapping cardinalities in ER-diagram?

(b) Construct an ER-diagram of a local medicine shop.

4+6

Please Turn Over

4. Consider the following relational schema :

Department (DepartmentID, DepartmentName)

Student (StudentID, Name, Age, CourseID)

Courses (CourseID, Name, Description, DepartmentID, Credit)

Perform the following using relational algebra.

- Find all students who have enrolled in courses having 4 credits.
 - Show the list of courses conducted under the department 'Computer Science'.
 - Give name of the student with highest age.
- 3+3+4
- Distinguish between Specialization and Generalization with the help of an example.
 - Describe any two data models of DBMS.
 - Explain logical data independence and physical data independence.
 - Define unary relationship with an example.
 - What is lossless join decomposition? How can you ensure that?
 - What is a 3NF relation? Give an example.
 - What are the problems of normalization?
 - Normalize the following relation to 2NF :

Student_Course

S_ID	S_Name	C_ID	C_Name	Credits
S1	Abhik Ray	C1	CC1	4
S1	Abhik Ray	C2	CC2	2
S2	Diya Basu	C1	CC1	4
S2	Diya Basu	C3	CC3	4
S3	Rahul Sinha	C2	CC2	2
S3	Rahul Sinha	C3	CC3	4

(2+1)+2+5

8. Write short notes on **any two** of the following :

5×2

- ANSI / SPARC Architecture
- Integrity Constraints
- Armstrong's Axiom
- Single-valued versus Multivalued Attributes.